PRODUCT DATA SHEET

DUROTILE UNIVERSAL C2S2

DUROTECH
Engineered Construction Chemicals
Made In Australia For Over 45 Years

FEATURES

- Durotile Universal C2S2 is a premium grade polymer modified, highly flexible, white cement based tile adhesive.
- Durotile Universal C2S2 is non slip/non slump and ideal for large format tiles.
- It is designed for bonding all types of ceramic, stone and mosaic tiles onto a variety of substrates like concrete, render, brickwork, blockwork, Gyprock, plasterboard, fibre cement and appropriately prepared flooring boards.
- It can be used internally or externally on wall and Floor surfaces.
- Durotile Universal C2S2 can be used for fixing Porcelain and Vitrified tiles.
- Durotile Universal C2S2 can be used to fix tiles over existing tiles as long as the existing tiles have been coated with Durotech Universal Primer.
- Durotile Universal C2S2 can be used to fix tiles over most waterproofing membranes. However it is advisable to contact the manufacturer prior to commencing.
- Durotile Universal C2S2 can be used for fixing tiles in fully immersed situations like swimming pools and spas.
- Durotile Universal C2S2 can be used over some acoustic mats please contact Durotech for more information.
- Durotile Universal C2S2 is suitable for tile installations over screeds or green screeds.
- Do not use Durotile Universal C2S2 for moisture sensitive stone like green marble.

TECHNICAL INFORMATION

Appearance	White Powder
Bulk Density	1.18 +/- 0.05
Open Time Approx	30 minutes @ 20°C
Adjustment Time	Approx 30 minutes @ 20°C
Pot Life	2 Hours @ 20°C
Ready for grouting	24 Hours @ 20°C
Light foot traffic	24 Hours
Ready for wet area service	3-4 days

TECHNICAL INFORMATION

- Durotile Universal C2S2 is suitable for use over 7 day old concrete which has a woodfloat finish.
- All rendered surfaces must be allowed to cure for at least 24 hours prior to commencing tiling.
- The maximum variation in the plane of the concrete must not exceed 5mm in 3 metres for floors and 4mm in 2 metres for walls.
- Steel trowelled finished concrete surfaces must be mechanically or abraded prior to commencing tiling.
- Fibre Cement sheet when used as an underlay must be a minimum of 5mm in thickness. For heavy duty commercial applications it should be a minimum of 9mm thick and all should be fixed in accordance with the manufacturers instructions and the relevant standards.
- Compressed Fibre-Cement sheets when used as a floor substrate must be 15mm thick and when used as a wall substrate must be 9mm thick and must be installed in accordance with the manufacturer's instructions and the relevant standards.

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- Gypsum - plasterboard sheets when used as a wall substrate must be a minimum of 10mm thick and installed in accordance with the manufacturer's instructions and the relevant standards.
- Ensure all surfaces are sound, dry and free from excessive movement, oil, dust, grease, wax, curing compounds, release agents and any other loose contaminating materials.
- All porous surfaces like concrete, screeds, should be primed using Duroprime.
- When applying the primer onto a floor surface it is recommended to firstly pour some primer in a section then spread the primer using a broom, brush or roller. Then continue this method of application until the entire area is primed. Note: This method of application ensures a thorough coat of the primer on the surface.
- Allow the primer to dry for approximately 30-40 minutes at 20ºC prior to commencing tiling.
- Any excess primer that has not dried should be removed with a rag prior to tiling.

EXPANSION I MOVEMENT JOINTS

Expansion / movement joints must be provided to allow for movement between adjacent building components. They should be as follows:
Over Existing joints in the substrate.

- Where two different substrates meet. e.g. Timber and Concrete.
- Around fixed elements in the floor e.g. Columns.
- At internal vertical corners.
- Around the perimeter of the Floor. ·In internal floors where any dimension exceeds 9m or 6m if subjected to sunlight. Framed floors should have an expansion joint every 4.5m In external floors where any dimension exceeds 4.5 m.
- On wall surfaces at storey heights horizontally and approximately 3m-4.5m apart vertically. Ideally they should be located over movement joints in the structural background and at structural material changes, for example the horizontal joint at the bottom of floor slabs, vertical joints at internal vertical corners, and at junctions with columns.
- Movement joints should go right through the tile adhesive bed to the background and be kept free from dirt and adhesive droppings. Movement joints must not be less than 6mm and not wider than 10mm.
- The movement joints must be installed as per AS3958:2007.

MIXING

- The mixing ratio of Durotile Universal C2S2 is 4.5 litres of water to each 15kg bag. Pour water into a clean drum. Gradually add the Durotile Universal C2S2 powder while mixing continuously until a smooth lump free mix is obtained.
- Always add powder to liquid. Allow to stand for 3 minutes, restir then apply the adhesive onto the substrate.
- The Open time of Durotile Universal C2S2 is Approx 30 minutes @ 20°C
- The Pot life of Durotile Universal C2S2 is 2 hours @ 20°C

COVERAGE

- A 15kg bag of Durotile Universal C2S2 will cover approximately 6-7m.

COVERAGE

- All tiling should be carried out in accordance with Australian Standard AS3958.1-2007.
- Once the surface has been appropriately prepared in accordance with Durotech's instructions then apply the adhesive onto the substrate using an appropriate notched trowel. For floor tiling use a 10mm x 10mm square notched trowel for tiles up to 300mm x 300mm. For tiles 300mm x 300mm and larger use a 12mm x 12mm square notched trowel and back butter each tile. For mosaic tiles use a 6mm x 6mm square notched trowel.
- For wall tiling use 6mm x 6mm square notched trowel for tiles up to 150mm x 150mm. For tiles larger than 150mm x 150mm use a 10mm x 10mm square notched trowel.
- Durotile Universal C2S2 should be applied onto the substrate at a rate of $1m^2$ at a time. Application rates greater than this can result in the tile adhesive skinning before the tiles are laid into it.
- Once the adhesive is applied onto the substrate ensure that it does not skin prior to bedding the tiles into it. Once the adhesive skins do not lay tiles into it, but remove it and apply fresh adhesive.
- When placing the tiles into the adhesive press them in by using a sliding action. Ensure no voids occur and full coverage of adhesive is under the tiles.
- For tiles with lugs, grooves or uneven backing it may be required to butter the back of the tile with adhesive in addition to trowelling the adhesive onto the substrate.
- The final bed thickness of the adhesive should be at least 2mm for wall tiling and 3mm for floor tiling.
- Once the tiling is completed do not disturb the tiled surface for at least 24 hours at 20ºC.
- Protect tiling from rain and inclement weather until 24 hours after grouting is complete.

GROUTING APPLICATION

- Grouting application can commence 24 hours after the completion of tiling.
- Grouting can be done using Durotech's range of grouts.
- Clean up can be done using a damp sponge.
- Ensure you use a clean sponge every time.
- Grout will fully dry in 24 hours at 20ºC, after which time the area can be put into service.

PACKAGING / SHELF LIFE:

- Durotile Universal C2S2 is available in 15kg bags.
- A bag of Durotile Universal C2S2, when stored in a cool, dry environment, and is stored above ground level, will have a shelf life of approximately 12 months.

HANDY TIPS

- Do not apply Durotile Universal C2S2 in temperatures above 40ºC and below 5ºC.
- Durotile Universal C2S2 cannot be used for fixing tile directly onto tongue and groove timber flooring.
- For applications / situations not mentioned in these instructions please contact your nearest Durotech office.
- Durotile Universal C2S2 being cement based is alkaline in nature, and therefore may cause dermatitis. It is recommended that applicators wear PVC gloves or similar and safety goggles.
- For a full SDS on this product please contact your nearest Durotech Office.

CLEAN UP

- Excess adhesive from the face of the tiles can be cleaned up with damp cloth while the adhesive is still wet
- Adhesive that has oozed out into the grout joint must be raked out with a knife / spatula etc.
- Tools and other equipment can be cleaned up using water while the adhesive is still wet

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